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2 April 1980

MILITARY SPECIFICATION

CAMERA, COPYING, LITHOGRAPHIC, DARKROOM,

MOBILE, PROCESS

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a mobile process copying camera.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	- Cushioning Material, Packaging (Cellulosic, Water Absorbent).
TT-L-20	- Lacquer, Camouflage.
TT-P-1757	- Primer, Coating, Zinc Chromate, Low Moisture Sensitivity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3610

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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture and Fungus-Resistant (for the Treatment of Communications, Electronic and Associated Electrical Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection).
MIL-T-704	- Treatment and Painting of Materiel.
MIL-F-3541	- Fitting, Lubrication.
MIL-C-3774	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-W-5086	- Wire, Electrical, 600-Volt, Copper, Aircraft.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52286	- Cable and Lead Assemblies, Power, Electrical, 300 to 600 Volts, 5 to 230 Amperes.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-150	- Photographic Lenses.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Government drawing. The following Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

ME

TA13222E5000 - Camera, Copying, Lithographic, 30 In. Lg. 24 In. Wide, Negative Size, 208V, 3 Phase, 60 Cycle.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JDT, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Std. No. 70 - The National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT (NMFTA)

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 2200 MILL Road, Alexandria, VA 22314.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT (SCAQMCD)

Rule 442 - Usage of Solvents.

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(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731).

UNIFORM FREIGHT CLASSIFICATION COMMITTEE, AGENT (UFCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The camera shall be in accordance with Drawing TA13222E5000, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the camera. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The camera shall be fabricated from compatible materials, inherently corrosion resistant or treated

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to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the camera may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the camera may be newly fabricated from recovered materials to the maximum extent practicable, provided the camera produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the camera.

3.4 Capacities, dimensions, and weight. The camera shall have the following capacities, dimensions, and weight:

Net weight: Maximum	3500 pounds
Operating height: Maximum	72 inches

Copyboard:

Opaque	36 x 48 inches
Transparent	21 x 27 inches

Range of reproduction:

19 inch lens	33 to 300 percent
24 inch lens	50 to 200 percent

Vacuum back:

Sensitized material

Minimum	4 x 5 inches
Maximum	24 x 30 inches

3.5 Rigidity. There shall be no observable shift of the projected image due to any sudden lifting of one corner of the camera skid or movement of personnel about the camera when the camera is mounted in a van-type mode. The skid

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platform shall be flat and true to line within 0.25 inch over its entire area when supporting the camera assembly, either on its wheels or on an uneven floor.

3.6 Vibration absorption. No movement or vibration of the projected image shall be observable when the vacuum exhausters are operating.

3.7 Shock absorption. When 10 foot-pounds of force are instantaneously applied upon the rear portion of the camera rail, the shock shall be dampened within 5 seconds after impact.

3.8 Accuracy. When supporting copy weighing up to 25 pounds, the camera shall reproduce, within its rated capacity of enlargement and reduction and within its field of lens coverage, all copy in accurate size and proportion to within 0.005 inch of the required dimensions. The photographic planes of the copyboard, transparency holder, lensboard, ground glass, and vacuum back shall be flat and parallel and shall maintain their parallelism throughout their range of movement to within 0.005 inch. This accuracy shall be retained following the cycle of positioning for transit; all hinges, doors, or counterbalanced parts shall function after movement in a van body across rough roads, and re-expansion of the camera into operating position. Shims may be used to obtain this accuracy.

3.8.1 Hand control. All hand actuated controls, knobs, wheels, and universal joints shall operate freely without binding.

3.9 Lead screws and counters. Lead screws for the camera drive controls shall be machined to 0.005-inch tolerance over any increment of their length. Counters shall be coupled to the lead screws and shall be geared to operate safely at speeds of not less than 7,500 figures per minute. Counters shall have visible figures and shall indicate the location of the lensboard and copyboard from the ground glass in increments of 0.01 inch.

3.10 Copyboard assembly. The copyboard and transparency holder assembly shall be pivotally mounted on the copyboard carriage so that both the copyboard and the transparency holder are held within 0.002 inch of the ideal photographic plane when in position for copying. The copyboard shall be equipped with vacuum parts arranged so that numerous sizes of copy can be held centered on the board. The surfaces of the copyboard and transparency holder shall be flat within 0.002 inch. Center lines and centrally located copy size scribe lines shall be permanently scribed on the surface of the copyboard. The center lines of the copyboard and transparency holder in the operating position shall be coincident with the optical axis of the lens. The operating surfaces of the copyboard and transparency holder shall be parallel with the vacuum back surface within 0.005 inch. The copyboard shall accept and hold in a flat plane any size of copy prepared on paper, sheet metal, or 0.375-inch plywood up to and including 36 by 48 inches. The transparency holder shall accept transparencies up to and including 21 x 27 inches.

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3.11 Copyboard illumination. The copyboard shall be illuminated by four solid state, pulsed xenon lamps. The lamps shall operate on a 208-volt, lamp controller and shall be equipped with an automatic voltage regulator.

3.12 Bellows. The bellows shall be of rubberized duck material to form a single accordion type bellows unit capable of contracting and extending with the full range of travel of the lens-board carriage.

3.13 Darkroom connecting frame and light barrier. The darkroom connecting frame shall provide a flexible light-tight joint between the camera back and the darkroom wall opening. The sliding curtains, flexible joint, the entire camera back, bellows fittings, and lens-board assembly shall provide a light-tight barrier between the camera room and darkroom.

3.14 Light-tightness. All openings in the camera back shall be provided with a light seal to insure that the entire camera back will be light-tight with no evidence of light leaks when viewed from the darkroom. Light seals shall be of a flexible material which will not transmit vibration to the camera.

3.15 Vacuum back. The perforated vacuum back surface shall have concentric rectangles permanently marked and clearly visible under darkroom illumination. The rectangles shall be centered about the optical axis and shall correspond in size to commercial standard film dimensions. The lines shall be fused into or absorbed by the surface so as to cause no surface distortion of the film. The holding face shall be plane and true to within 0.002 inch and shall have no surface irregularities of a size that will result in local distortion of the film negative. The porosity of the vacuum back and the suction of the pump shall be sufficient to hold any size of film up to the rated capacity of the camera, through zone control only, without the aid of additional mechanical devices. The emulsion surface of the film mounted on the vacuum back shall occupy to within 0.002 inch the same photographic plane of the front surface of the ground glass when the vacuum back door is locked into exposure position.

3.16 Ground glass. The ground glass shall be pivotally mounted so that it may be positioned directly in line with the lens for alinement and focus of the copy image or swing out of the way of the vacuum board. The front surface must be flat within 0.002 inch. When in position for alinement and focus, it must be parallel with the copyboard within 0.005 inch, and be located within 0.002 inch of the focal plane of the vacuum back. The grained surface of the ground glass shall resolve not less than 20 lines per millimeter in a projected image. A series of rectangular shaped grid lines shall be scribed in the surface and accurately oriented about the center point of the glass.

3.17 Focusing system. The camera shall be calibrated to a predetermined focusing system for positioning the lens-board and copyboard by charted readings in 1 percent increments for any ratio of enlargement or reduction over the specified ranges of the lenses furnished. Such positioning shall provide the

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maximum resolving power for the lenses, as well as accuracy of both horizontal and vertical image dimensions on the ground glass, across the maximum rated field of coverage for all ratios of reproduction.

3.18 Focusing charts. Three copies of a focusing chart shall be provided for each lens with legible counter-readings for each percentage of reproduction and enlargement over the specified range of reproduction (see 3.4). The focusing charts shall include the following information: Lens make, serial number, nominal focal length, nominal maximum aperture, back focal distance, equivalent focal length, nodal point separation, lens thickness, and camera serial number. The charts shall be approximately 9 by 12 inches and shall be laminated in transparent plastic.

3.19 Test chart. A metal test chart for checking the calibration and accuracy of the camera shall be provided in a protective container. The test chart shall consist of a hard aluminum alloy sheet not less than 0.032 inch thick on which a white lacquered or enameled surface forms the base for a photographically transferred image from the master grid used in preparing the ground glass. The grid lines shall be spaced at right angles within individual and overall tolerances of 0.003 inch. The image shall be given a hard finish by heat treatment or other protective means. The image on the test chart shall be made so that conjugate lines and numbers will match when the test target is placed on the copyboard and viewed through the ground glass at a ratio of 1:1.

3.20 Lenses. Two interchangeable, color-corrected, process lenses conforming to MIL-STD-150, type IV, shall be provided with each camera. The nominal focal length of the lenses shall be 19 inches and 24 inches respectively, and the relative aperture shall be f/11. The lenses may have a larger relative aperture; however, the range shall be between f/11 and f/64. The maximum variation of focal lengths when measured photographically through Wratten A, B, or C5 filters over the visible and near ultraviolet spectral range shall be not more than 0.2 percent. The half-angular field of coverage of the lenses shall be not less than 22 degrees for both lenses. Each lens shall be mounted inside a barrel which is mounted on a lens-mount plate conforming to the drawings. Aperture markings shall be visible when observed from the right side of the camera while facing the copyboard. Each lens shall be marked with the manufacturer's name, lens name, serial number, nominal focal length in inches, and maximum aperture. Lenses shall withstand 155 ^{o}F $\pm$2 ^{o}F for not less than four hours and -65 ^{o}F $\pm$2 ^{o}F for not less than twelve hours without evidence of damage such as cracking, chipping, or breaking.

3.21 Lens coating. All air-glass surfaces of each lens shall be coated with a reflection-reducing coating conforming to MIL-C-675.

3.22 Lens distortion. The distortion in either lens at f/11 aperture shall not exceed that which would cause a point in the image plane to be displaced from its theoretically distortion-free position by more than 0.05 mm at a ratio of 1:1 reproduction throughout an angular field of 0 to 22 degrees.

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3.23 Diaphragm aperture control. Provisions shall be incorporated to correlate the f-stop openings to the ratio of reproduction for equivalent light transmission of the lenses throughout the range of reproduction specified in 3.4. The iris diaphragm shall be graduated and marked as follows: f/11, f/16, f/22, f/32, f/45, and f/64. It shall be capable of incremental intermediate stops between the major stops.

3.24 Resolving power. The resolving power of the lenses in terms of lines per millimeter (mm) in the image plane shall be not less than the values shown in table I.

TABLE I. Resolving power.

Angle from center	24-inch lens	19-inch lens
(Degrees)	(Lines per mm)	(Lines per mm)
0	23	25
5	20	25
11	20	25
16	12	22
22	12	4

3.25 Lens accessories. The following set of accessories shall be provided for each lens: a lens cap; a lens box; seven interchangeable filter frames; a dust-tight filter case with slots to accommodate the filter frames; and a guide detailing use and care of the lenses. Both sets of accessories shall be enclosed in a felt-lined protective compartmentized box having a hinged cover and lock.

3.26 Electrical requirements. The camera shall operate on 208V, 3 phase, 60 Hertz alternating current electrical supply. The electrical equipment shall be as shown on the drawings. Motor receptacle connector, cable assembly, and dummy connector shall conform to applicable portions of MIL-C-52286, and figures 1 through 6 therein. The electrical wire shall conform to MIL-W-5086. All electrical equipment and wiring shall conform to NFPA No. 70. Helical, spring-wound, rubber-covered cord shall be used from hinged duct to camera lens-board and copyboard. All wiring shall be color coded and circuits numbered. A plastic-enclosed or an etched metal wiring diagram detailing color coded wires and identifying switches and controls by name and part number shall be mounted on an electrical box or other suitable location.

3.27 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.27.1 Lubricants. The camera shall operate as specified herein when lubricated with military lubricants (see 6.4).

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3.27.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859, type I. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand operable, quick-disconnect fasteners may be removed to provide accessibility.

3.27.3 Delivery lubrication service. When delivered, all components requiring lubrication shall be lubricated. Grease lubrication, including lubrication of sealed bearings, shall be with standard military thermally stable greases conforming to MIL-G-10924 or MIL-G-23827, as applicable. Each item of equipment shall be tagged in a conspicuous place to indicate the temperature range and grades of greases used.

3.27.4 Lubrication chart. A lubrication chart of permanent construction shall be mounted at the operator station of the camera.

3.28 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with varnish conforming to MIL-V-173, composition I or II as applicable (see 3.28.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.28.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall make available to the Government evidence that the use of composition II is required, and shall make available certification that the composition II material complies with Rule 442, South Coast Air Quality Management Control District.

3.29 Instruction plates. Each camera shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the camera. Instruction plates shall conform to MIL-P-514, type III.

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3.30 Treatment and painting. The portions of the camera and its components normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. All parts of the camera which may reflect light into the lens or onto the film shall be cleaned and treated in accordance with MIL-T-704, given a primer coat conforming to TT-P-1757 and a finish coat conforming to TT-L-20, color black.

3.31 Storage. Camera and all ancillary equipment and spare parts shall withstand prolonged storage at temperatures of -65 °F to 155 °F and relative humidities ranging from 10 percent to 90 percent without degrading operation of camera when returned to normal operating conditions (see 4.5.2.9).

3.32 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantity indicated will be loaned by the Government (see 6.3).

Item No.	Description	Quantity
1.	Map reproduction van body, semitrailer mounted.	1 ea.

3.33 Safety.

3.33.1 Exposed parts. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All fragile components, such as transparency holder glass, ground glass, xenon lights, and flash lamp shall be mounted and guarded to prevent breakage and damage during use and transit. Protective devices shall not impair the operating functions.

3.33.2 Operations. All operations of the camera are to be designed to be safe so as to avoid hazards to any operating or transient personnel in the vicinity of the camera.

3.34 Human factors engineering. The camera shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited, to general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainability (5.9), and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.35 Workmanship. All parts, components, and assemblies of the camera including forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, pits, scale, and other harmful extraneous material. All exposed edges which could cause injury to an operator shall be rounded or chamfered.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article camera shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article camera shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each camera shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each camera shall be tested as specified in 4.5.2.1 and 4.5.2.3 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cameras shall be examined as specified herein for the following defects:

101. Nonconformance to drawings (see 3.1.1).
102. Material not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).
104. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.2).
105. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.3).
106. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the camera (see 3.3.4).
107. Capacities, dimensions, and weight not as specified (see 3.4).
108. Skid platform not flat and true to line within 0.25 inch as specified (see 3.5).
109. Hinges, doors, or counterbalanced parts fail to operate, after movement of camera (see 3.8).
110. Manual drive handles or other operating controls bind (see 3.8.1).
111. Copyboard and transparency holder not within 0.002 inch of ideal photographic plane as specified (see 3.10).
112. Copyboard not properly illuminated, four solid state pulsed xenon lamps not functioning (see 3.11).
113. Perforated vacuum back markings not as specified (see 3.15).
114. Vacuum back and suction of the pump not as specified (see 3.15).
115. Counter readings of focusing charts not as specified (see 3.18).
116. Aperture and lens markings not as specified (see 3.20 and 3.23).
117. Lens coating not as specified (see 3.21).
118. Lens accessory box improperly assembled (see 3.25).
119. All accessories not protected from damage in lens accessory box as specified (see 3.25).
120. Cable assembly not as specified (see 3.26).
121. Electrical equipment not operating (see 3.26).
122. Lubricants and lubrication fittings not as specified (see 3.27.1, 3.27.2, and 3.27.3).
123. Lubrication chart illegible or absent (3.27.4).
124. Electrical circuitry not treated to retard fungus growth (see 3.28).
125. Fungus and moisture resistance not as specified (see 3.28).
126. Instruction plates, diagrams illegible or missing (see 3.29).
127. Treatment and painting not as specified (see 3.30).
128. Protective devices non-functional or absent (see 3.33).
129. Human factors engineering not as specified (see 3.34).
130. Workmanship not as specified (see 3.35).

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4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, those parts of the camera normally requiring lubrication shall be lubricated with oils and greases specified in 3.27.

4.5.2.2 Rigidity, vibration absorption, and shock absorption. The camera shall be mounted in the Government-loaned map reproduction van body, semitrailer mounted (see 3.32).

4.5.2.2.1 Rigidity. Printed copy with halftone dots shall be placed on the copyboard of the camera, and the projected image shall be observed with a 40 power magnifying glass while two or more individuals walk about the van. One corner of the camera shall be lifted 0.50 inch above the floor while the projected image is being viewed on the ground glass. Any observable image shift, as specified in 3.5, shall constitute failure of this test.

4.5.2.2.2 Vibration absorption. With the vacuum exhausters in operation, the projected image shall be observed with the 40 power magnifying glass. Any observable image shift, as specified in 3.6, shall constitute failure of this test.

4.5.2.2.3 Shock absorption. A 10-pound lead weight shall be dropped from a height of 12 inches upon the rear portion of the camera rail while the projected image is being observed on the ground glass. Incomplete dampening within 5 seconds after impact, as specified in 3.7, shall constitute failure of this test.

4.5.2.3 Accuracy. The camera shall be tested optically and mechanically for accuracy. The optical test shall utilize the focusing system, the test chart, and the ground glass. Tests shall be conducted on a minimum of 5 ratios of enlargements and reduction on each lens, and shall include the ratio 1:1 and the maximum and minimum positions covered by each lens. The projected image shall be compared to the size of the test chart to determine the size of enlargement or reduction. Mechanical tests shall be conducted with a precision trammel equipped with a dial indicator. Tests shall include measurements to determine parallelism of planes and the ability of the camera controls to position the lens board and the copyboard for any desired enlargement or reduction and to repeat these positions. During this test, the location and functioning of the limit switches and stops on the camera ways and carriages shall be tested to ensure that they are properly positioned to prevent excessive movement or interference between the carriages when the camera is power driven or hand operated. Nonconformance to 3.8, 3.9, 3.10, 3.15, 3.16, 3.17, or 3.19, or evidence of excessive movement or interference between the carriages when the camera is power driven or hand operated shall constitute failure of this test.

4.5.2.4 Resolving power. Each lens shall be tested individually at a magnification of 1:1, at an aperture setting of f/22. The resolving power tests shall be conducted using targets and target layout in accordance with

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MIL-STD-150. Test targets shall be of high-contrast type with dark lines on a light background; illumination shall be by solid state, pulsed xenon camera lights with a total power of 6 kW. Nonconformance to 3.24 shall constitute failure of this test.

4.5.2.5 Lens distortion. The lens distortion shall be tested visually in accordance with MIL-STD-150, method 30, with the lens mounted on the camera at a reproduction ratio of 1:1. The test shall be made with the lens at maximum aperture and the light source shall be the solid state, pulsed xenon camera lamps with a total power of 6 kW. A point in the image plane that is displaced from its theoretically distortion-free position by more than 0.05 mm at a reproduction ratio of 1:1 throughout an angular field of 0 to 22 degrees shall constitute failure of this test (see 3.22).

4.5.2.6 Light-tightness. The camera shall be set up for operation by positioning it in a wall opening of the required size with the rear of the camera in the darkroom. The bellows shall be extended full length and the shutter shall be closed. All light in the darkroom shall be extinguished, and the solid state, pulsed xenon lamps in the camera room shall be directed toward the copyboard. The inspector shall remain in the darkened darkroom not less than 5 minutes prior to conducting the test to allow the inspector's eyes to adjust to the darkness. The darkroom wall and bellows shall then be observed for light leaks. Nonconformance to 3.12, 3.13, and 3.14 and evidence of any light leaks shall constitute failure of this test.

4.5.2.7 Vacuum system. The vacuum back, vacuum copyboard, and transparency holder shall be tested for adherence by using a range of film sizes from 4 by 5 inches to 24 by 30 inches on the vacuum back and transparency holder, and copy sizes from 9 by 9 inches to 36 by 48 inches on the vacuum copyboard to assure satisfactory adherence. The film to be used for this test shall be the same as used for the resolution test in 4.5.2.4. Film may be used to represent copy sizes from 9 by 9 inches to 24 by 30 inches. The maximum copy sheet size shall be tested with a 36 by 48 by 0.375-inch plyboard. Lack of adherence of any sheet or board, as specified in 3.15, shall constitute failure of this test.

4.5.2.8 Vibration resistance. The camera shall be mounted in the Government-loaned semi-trailer mounted van body and shall be positioned for transit. The complete vehicle with the camera mounted therein shall be driven not less than 20 miles, of which not less than 10 miles shall be unsurfaced, stabilized, or partially surfaced earth or gravel roads. The test shall include ten abrupt starts, ten abrupt stops, and five sharp turns at maximum safe speeds in each direction (right and left). The route shall be approved by the contracting officer. At the completion of the test run, the camera shall be operated. Any evidence of damage to the camera or non-conformance to 3.8, 3.9 3.10, 3.15, 3.16, and 3.17 shall constitute failure of this test.

4.5.2.9 Storage. The camera and all ancillary equipment and tools shall be placed in a temperature test chamber at normal room temperature. The temperature of the chamber shall then be raised to +155 ^{o}F $\pm$2 ^{o}F and maintained at this

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temperature for not less than 12 hours. The chamber shall then be returned to normal room temperature and the camera and equipment shall be examined for damage and operated to verify the performance in accordance with 4.5.2.3, 4.5.2.4, 4.5.2.5, 4.5.2.6, and 4.5.2.7; nonconformance in any of the above and nonconformance to 3.31 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.6.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of production.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects:

131.	Preservation method not as specified for level A or C (see 5.2.1.1 and 5.2.2).
132.	Movable components of cameras not positioned to realize minimum cube and the components not secured to prevent movement as specified for level A (see 5.2.1.1).
133.	Lamps not cushioned and placed in box as specified for level A (see 5.2.1.2).
134.	Lenses not preserved as specified for level A (see 5.2.1.3).
135.	Repair parts and accessories not preserved as specified for level A (see 5.2.1.4).
136.	Maintenance tools not preserved in accordance with referenced document as specified for level A (see 5.2.1.5).
137.	Technical publications not preserved as specified for level A (see 5.2.1.6).
138.	Consolidation not as specified for level A or C (see 5.2.1.7 and 5.2.2).
139.	Shipping containers not as specified for level A or B (see 5.3.1 and 5.3.2).
140.	Blocking and bracing not as specified by the referenced document for level A or B (see 5.3.1 and 5.3.2).
141.	Strapping not as specified for level A or B (see 5.3.1 and 5.3.2).
142.	Marking illegible, incomplete, or incorrect for level A, B, or C (see 5.4).

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5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack within the time frame specified to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.2).

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cameras. Each completely assembled camera that can be secured to the crate base specified in 5.3.1 and 5.3.2, shall be preserved in accordance with MIL-P-116, method IIa. The camera shall be secured by bolting directly through the barrier material to the crate base. Movable components shall be positioned to realize minimum cube, and the positioned components secured to prevent movement within the barrier material.

5.2.1.2 Xenon lamps. Each xenon lamp shall be removed from the copyboard and completely wrapped with cushioning material and placed in an individual close-fitting fiberboard box conforming to PPP-B-636, type weather resistant.

5.2.1.3 Lenses. Lenses with their mounts and caps in place, contained in the boxes provided, shall be preserved in accordance with MIL-P-116, method IIb. The lens boxes shall serve as the inner containers. The lens boxes containing the lenses shall be completely wrapped with a minimum of 1 inch thickness of cushioning material conforming to A-A-1898, grade II, style optional, secured in place with tape conforming to PPP-T-60, type IV, prior to placing into the method II barrier material. The outer container shall be a box conforming to PPP-B-636, W5c or W6c, style RSC. The outer container shall be closed in accordance with method V of the appendix to the box specification.

5.2.1.4 Repair parts and accessories. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts and accessories.

5.2.1.5 Maintenance tools. Maintenance tools for each camera shall be preserved and packaged in accordance with PPP-P-40, level A.

5.2.1.6 Technical publications. The technical publications for each camera shall be preserved together in accordance with MIL-P-116, method IC-1.

5.2.1.7 Consolidation. The preserved repair parts, lamps, lenses, accessories, maintenance tools, and technical publications for each assembled camera shall be consolidated in close-fitting boxes conforming to PPP-B-601, overseas type, grade B. Cushioning, blocking, and bracing shall be provided to prevent movement of the contents within the boxes. Strapping shall not be required.

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5.2.2 Level C. Each complete camera shall be preserved as specified for level A except method IA-16 shall be used in lieu of method IIa and all containers shall be domestic type or class, whichever is applicable.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Each camera, and the consolidated components, preserved as specified in 5.2, shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style A. Strapping shall conform to ASTM 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675. Blocking, bracing, and anchoring of the crate contents shall be in accordance with the crate specification and MIL-STD-1186.

5.3.2 Level B. Each camera, and the components preserved as specified in 5.2, shall be packed as specified for level A except the crate shall conform to MIL-C-3774, type I style A and strapping may be coated finished.

5.3.3 Level C. Each camera, and the consolidated components, preserved as specified in 5.2, shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The camera is intended to be used in mobile vans for the precision copying of maps and other printed matter for photolithographic purposes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issues of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Time frame required for submission of the first article model (see 3.2).
d. When van body is not Government-loaned (see 3.32).
e. Level of preservation and packing required (see 5.2 and 5.3).
f. Any special marking (see 5.4).

6.3 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.32.

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6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the camera covered by this specification as contained in Federal Supply Catalogue, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the cameras in accordance with the specification.

6.7 Replacement parts and tools. The following replacement parts and tools should be furnished with the camera:

a. One ground glass (focusing).
b. One ground glass for transparency holder.
c. One set of trammels with dial indicator.
d. Four xenon bulbs.
e. Oiler, hand, 1/2 pint capacity, NSN 4930-00-537-8977 - 1 ea.

6.8 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions regarding shipment of camera.

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6.10 Subject term (key word) listing.

Copyboard
Diaphragm aperture control
Focusing charts
Vacuum back

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3610-A363

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-52961A(ME)	2. DOCUMENT DATE (YYMMDD) 92/04/02

3. DOCUMENT TITLE
Camera, Copying, Lithographic, Darkroom, Mobile, Process

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3467	(2) AUTOVON 654-3467
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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